Silica: Nature's Hidden Shield Protecting Plants from Insect Pests

R Saranya*[1] and P Sikala[2]

[1*,2]Department of Agricultural Entomology, Tamil Nadu Agricultural University, Coimbatore – 641003, Tamil Nadu, India.

[1*]Corresponding author E-mail: saranyaramamoorthy30@gmail.com

Abstract

Insect pests cause substantial yield losses in agricultural crops, forcing farmers to rely heavily on chemical pesticides. Recent research highlights the role of silica as a natural, eco-friendly defense mechanism in plants. Silica strengthens plant tissues, disrupts insect feeding and enhances internal defense responses, thereby reducing pest damage without harming the environment. This article explains how silica-based plant defense works and why it holds promise for sustainable pest management.

Introduction

Agriculture today faces a dual challenge, protecting crops from insect pests while reducing the negative impacts of chemical pesticides. Insects damage crops at various stages, affecting both yield and quality. Although pesticides provide quick control, their overuse has led to problems such as pest resistance, residue accumulation and environmental pollution.

Interestingly, plants are not helpless. They possess natural defense strategies that help them withstand insect attack. One such powerful but often overlooked defense is silica, a naturally occurring compound that enhances plant resistance to pests. Understanding and utilizing this mechanism can support safer and more sustainable crop protection.

What Is Silica and Why Does It Matter?

Silica is derived from silicon, the second most abundant elements in the Earth's crust. Plants absorb silicon from the soil in the form of silicic acid and deposit it as solid silica in their tissues. These deposits, known as phytoliths, act like tiny shields within plant cells(Figure 1).

Although silicon is not considered an essential plant nutrient, its beneficial role in improving plant strength, stress tolerance and pest resistance is now well recognized. Also, due to this function silicon is recognised as a quasi-essential element.

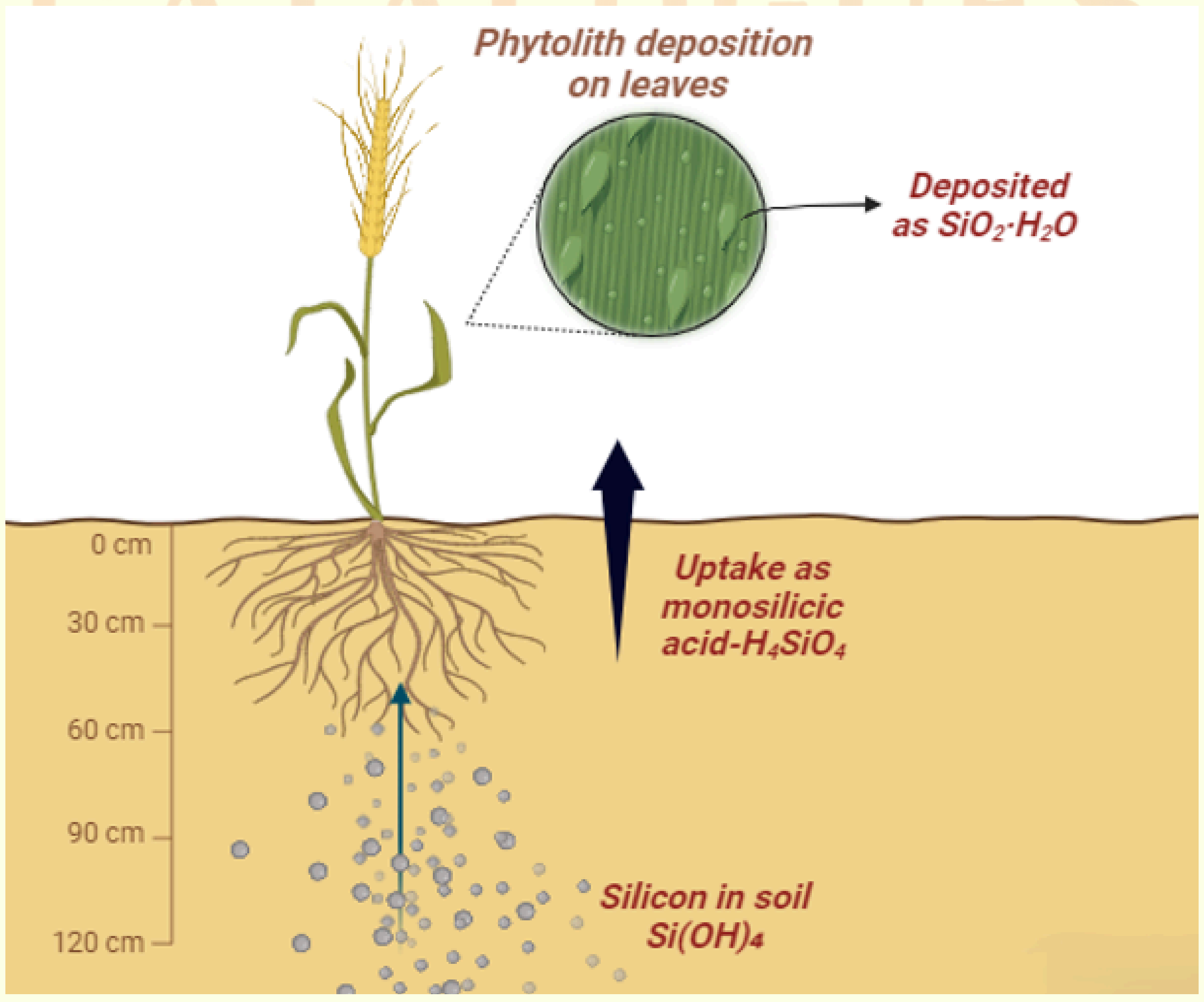

Figure. 1 Mechanism of phytolith formation

Silica-based defense operates through multiple, complementary mechanisms:

- **Physical protection:** Silica accumulation makes leaves, stems and fruits harder and tougher. This makes it difficult for insects to chew or bore into plant tissues, reducing feeding damage.
- **Reduced insect performance:** Insects feeding on silica-rich plants experience poor digestion and faster wear of mouthparts, leading to slower growth and reduced survival.
- **Enhanced internal defenses:** Silica activates plant defense pathways, increasing the production of protective enzymes and compounds that deter insects.
- **Lower pest preference:** Silica-treated plants are often less attractive to insects, reducing infestation levels.

Together, these mechanisms synergistically make plants more resistant rather than directly killing pests.

Evidence from Agricultural Crops

Studies across cereals, sugarcane and vegetables have shown that silica application significantly reduces damage from chewing and boring insect pests. In crops such as rice, brinjal and maize, silicon supplementation has resulted in lower pest incidence, improved plant vigour and better yield quality. Importantly, silica strengthens the plant's own defense system, making protection more durable and less prone to pest resistance.

Silica as a Sustainable Pest Management Tool

One of the greatest advantages of silica-based defense is its environmental safety. Silica is non-toxic to humans, animals and beneficial insects when used correctly. By reducing reliance on chemical pesticides, silica supports biodiversity and food safety.

Silica fits well into Integrated Pest Management (IPM) programs, where it can be combined with biological control, cultural practices and need-based pesticide use.

How Can Silica Be Applied in Farming?

Farmers can use silica through:

- Soil application of silicon fertilizers
- Foliar sprays of soluble silicon formulations
- Organic sources such as rice husk ash

Timely application during key crop growth stages ensures better uptake and effective pest resistance.

Conclusion

Silica-based plant defense represents a shift from killing pests to strengthening plants. By reinforcing plant tissues, disrupting insect feeding and activating internal defense responses, silica acts as a natural shield against insect pests. Its eco-friendly nature and compatibility with sustainable farming practices make it a promising tool for future agriculture. Harnessing silica not only protects crops but also supports safer food production and environmental health.

References

Epstein, E. (1999). Silicon. Annual Review of Plant Physiology and Plant Molecular Biology, 50, 641–664.

Ma, J. F., & Yamaji, N. (2006). Silicon uptake and accumulation in higher plants. Trends in Plant Science, 11(8), 392–397.

Reynolds, O. L., Keeping, M. G., & Meyer, J. H. (2009). Silicon-augmented resistance of plants to herbivorous insects: A review. Annals of Applied Biology, 155(2), 171–186.

Massey, F. P., Ennos, A. R., & Hartley, S. E. (2006). Silica in grasses as a defence against insect herbivores. Proceedings of the Royal Society B: Biological Sciences, 273(1592), 229–235.

Saranya, R., Suganthy, M., Ganesan, K., Rajkishore, S. K., Sathiya Bama, K., Janaki, P., & Priya Varshini, A. C. (2024). Silica Shield: Harnessing phytoliths for sustainable plant protection. Silicon, 16, 5771–5789.

Agri
CATALOGUES